MUHAMMAD SHOAIB

AI Research Student & Developer | BS Artificial Intelligence, Aror University Sukkur

Sukkur, Sindh, Pakistan
+92 316 3813033
mshoaibb707@gmail.com
www.linkedin.com/in/muhammad-shoaib/
github.com/MShoaib-81
https://shoaibb.vercel.app

PROFILE SUMMARY

Motivated AI and Machine Learning student with a strong foundation in Deep Learning, medical image analysis, and intelligent systems. Passionate about applying ML/DL techniques in healthcare, mental health detection, and explainable AI. Actively developing research-driven projects in computer vision and natural language processing to build impactful, real-world solutions.

RESEARCH INTEREST

Machine Learning and Deep Learning with applications across Computer Vision, Natural Language Processing, and Speech Technologies, with a strong focus on Medical AI and Mental Health Tech to develop socially impactful, ethical, and explainable AI systems.

EDUCATION

BS Artificial Intelligence **2023– Ongoing**
Aror University Sukkur

Intermediate In Science **2020 - 2022**
Islamia Science College Sukkur

TECHNICAL SKILLS

Programming: Python, C++, Java, HTML/CSS, MySQL
AI/ML: TensorFlow, Keras, Scikit-learn, OpenCV, Rasa
Tools: Git/GitHub, Jupyter, VS Code, Anaconda, PowerBI, Figma

CERTIFICATIONS

- Google AI Essentials
- Google Cyber Security
- C++ Fundamentals
- Python Programming
- Web Development
- Data Analysis

RECOMMENDATIONS

Dr Wazir Ali
Faculty of Emerging Sciences and Technology Assistant Professor & Incharge: Department of Artificial Intelligence
wazir.faculty@aror.edu.pk

Dr Irfan Ali
Faculty of Emerging Sciences and Technology Assistant Professor & Incharge: Department of Multimedia & Gaming
Irfan.faculty@aror.edu.pk

Mr Abdul Aziz
Faculty of Emerging Sciences and Technology Lecturer, Department of Artificial Intelligence
aziz.faculty@aror.edu.pk

RESEARCH & PROJECTS

***Beyond Accuracy: Explainability of Transformer Models* (CSET'26)**
Published research on low-resource NLP at CSET'26 (QUEST).

- Developed a fine-tuned UrduBERT model capable of 79.8% F1-score, utilizing a hybrid attribution pipeline to solve model opacity issues.

Depression Detection on Social Media **Ongoing**

- Developing an NLP-based system to detect depression from social media text using ML and deep learning.
- Implementing data collection, preprocessing, feature extraction, and classification models.

Sindhi Text-to-Speech (TTS) System

- Built a Sindhi TTS system using baseline models and a transformer-based approach (SpeechT5) for natural and clear speech synthesis, focusing on accessibility and language inclusion.

Facial Emotion detection

- Designed a CNN-based deep learning system for classifying emotions from facial expressions. Achieved high accuracy on benchmark datasets and explored real-time deployment for human-computer interaction.

Sign Language Recognition

- Built a computer vision model to recognize hand gestures for sign language interpretation. Applied CNNs and image preprocessing techniques to improve classification performance.

Fake News Detection

- Implemented NLP-based models (TF-IDF, Word Embeddings, BERT) to detect fake news articles. Focused on improving generalization and robustness against misinformation.

Advanced Regression Models

- Explored and implemented Linear, Polynomial, Lasso, and Ridge regression techniques for multivariate analysis. Conducted performance evaluations and comparisons with real-world datasets.

SOFT SKILLS

Problem Solving • Leadership • Communication • Critical Thinking • Teamwork • Research

AVAILABILITY

- Research Internships
- AI/ML Collaborations
- Technical Consulting
- Speaking Engagements
- Academic Research Partnerships